Question 1

The value of a real estate investment is driven by risks and opportunities.

a- What are the main risk and opportunities of real estate investment? (10 points)

These issues were discussed in Lecture 1.

b- What makes real estate different from other financial assets? (5 points)

These issues were discussed in Lecture 1.

c- How does a discounted cash flow (DCF) allow you to account for risks and opportunities in assessing the value of a real estate investment opportunity? (5 points)

These issues were discussed in Lectures 1 and 2.

d- Discuss the key elements you want to consider in your operation pro-forma to estimate the net operating income (NOI) of a property. Add some insights you gained from working at the two case studies. (15 points)

There was an extensive discussion of these issues in Lecture 2. The students also engaged with two Case Studies and should report on what they learnt from these two exercises in terms of NOI estimation.

e- How do you determine the re-sale value of a real estate investment? (10 points)

Same as above. The students had plenty of opportunities to think about these issues because of what was discussed in Lecture 2 and the two case studies.

f- 'Where' and how do you consider sustainability and environmental issues in the calculations you carry out to determine a property's NOI (discussed at point d-) and its sale value (discussed at point f)? (15 points)

We hosted a roundtable on sustainability in real estate in Week 10. We also touched upon these issues in the discussions of the two case studies and at various other times during lectures (e.g., lectures 1 and 2).

For-resits

g- How does investing in REITs allows to by-pass some of the key risks of direct real estate investment? (10 points)

These issues were discussed last year in the introductory part of the lecture on REITs.

h- What are the key constraints that affect how REITs operate? And what are the key advantages – for both investors and real estate asset managers? (15 points)

These issues were discussed last year in the Lecture on REITs both by the regular lecturer (Prof. Olmo Silva – via a pre-recorded lecture) and by an external speaker (Mr. Francis Salway – former CEO of one of the biggest UK REITs).

i- What key features would you consider when constructing a portfolio of indirect real estate investments? (15 points)

These issues were discussed by an external speaker (Dr. Niko Szumilo) who delivered the last lecture – focussed on direct/indirect real estate portfolio construction. The students here are asked to focus on indirect real estate investment – which follows from the discussion on REITs.

For current students

a. Real estate investment often entails land development rather than simple building acquisition and management. Why is this route into direct real estate investment riskier than just buying and operating a building? What risks do you need to keep in mind? Provide some insights you gained from working at the first case study. (10 points)

These issues are new to the modules for the current academic year. They were discussed at the start of Lecture 4.

b. Discuss the relative roles and structuring of interim and permanent finance (loans) in real estate development. (15 points)

These issues were discussed in the new Lecture 4 as well as in the first case study (presented in week 8).

c. A quick assessment of the profitability of a development project can be obtained by using the 'build to a 10, sell to an 8' rule (Linneman, Ch. 9). Explain what this statement means and provide a simple numerical example. (15 points)

These issues were discussed in the new Lecture 4.

Question 2

You are planning to buy a property worth £1,000,000. You have a deposit of 750,000 and a (net) annual income of £57,500. You are considering a variety of mortgages. All of them have monthly compounding and 20 years maturity. The bank you are dealing with uses a debt coverage ratio of 1.5 and an income multiple of 4.5.

a- Consider an interest-only (IO) loan. The bank charges an annual interest rate of 12%. What is the value of the monthly payments you owe to the bank? Do you qualify for this loan based on the DCR and income multiple rules? (5 points)

The loan of 250,000 would be charged a interest of 12%/12=1%. The monthly payment would therefore be 2,500. The person's monthly income is 57,500/12 = 4,792. So, the person qualifies on the bases of the DCR (4,792 / 2,500). The person also qualifies on the basis of the income multiple – just! – as they are borrowing 250,000 and could borrow up to 57,500 * 4.5 = 258,7500. Such calculations were discussed at several points in lectures 3 and 6.

b- Consider instead a constant payment mortgage (CPM). The loan charges an annual interest of 10%. What is the value of the monthly instalments you need to pay to the bank? Do you qualify for this loan? (5 points)

Constant payment mortgage calculations were discussed during lecture 6 and 8. In this case, the students need to set compounding to 12 (monthly) and enter data as follows: N=20 (since there is compounding at 12, this is transformed into 240), I/Y=10% (the calculator automatically translates into the monthly equivalent); PV=-250,000. The monthly PMT of this loan is 2,412.55. The person still qualifies for the loan as the PMT are lower than in the case of the IO loan.

c- Why does the bank charge a lower interest on the CPM than on the IO loan? What do banks 'like' in a CPM loan compared to an IO one? And do borrowers like / dislike about IO and CPM loans? (10 points)

These issues were extensively discussed in Lecture 6.

d- The bank has also informed you that both loans carry a discount point of 1% and a pre-payment penalty charge of 2%. What is the effective cost of borrowing for the IO loan if you reimburse the loan at maturity? And what is the effective cost of borrowing for the CPM loan if you repay at maturity? Compare your findings and discuss what you notice. (10 points)

The IO carries an effective cost of borrowing of 12.13% if held to maturity. This comes from 240 payments at 2500 and a final reimbursement of 250,000 vis a vis an effective loan of 247,500 (1% discount point). For the CPM, the calculations are 247,500 loan vis a vis payments of 2412.55 throughout and a zero outstanding balance at the end. The effective cost of borrowing is: 10.14%. This is a bigger increase (on a 10% basis) than the increase for the IO (on a 12% basis) because the extra costs for the CPM are front loaded. Note that the pre-payment penalty here is a red-herring but it's useful for the next exercise.

e- What are the effective costs of borrowing for the CPM and IO loan if instead you decide to repay it after 10 years? And if you repay it after 5 years? What do you notice? (10 points)

Now the IO repayment after 10 years (120 periods) or 5 years (60 periods) incurs a 2% charge bringing the total to 255,000. So if the borrower repays after 10 years, the calculations are 247500 vis a vis 119 periods of 2500 and 1 period of 2500 + 255000. This gives an IRR of 12.277%. If repayment occurs after 5 years, then the payments are 59 periods of 2500 and 1 period of 2500 + 255,000. This gives an IRR of 12.56%. The sooner you repay, the higher the cost of borrowing. This is expected.

For the CPM, one has to work out the outstanding amounts for the loan after 10 years (120 period) and 5 years (60 periods). These are respectively, 182560.8 and 224,506.1. On top of these, there is a 2% charge so the outstanding balances are 186,212 and 228,996.2. Using these numbers, one can calculate the IRR of loan with the two horizons as follows. 10 years: 247500 loan against 119 payments of 2412.55 and 1 payment of 2412.55 + 186,212. IRR = 10.27%. 5 years: 247500 loan against 59 payments of 2412.55 and 1 payment of 2412.55 + 228,996. IRR = 10.56%.

f- What is the rationale for banks to impose origination charges? (5 points)

These issues were extensively discussed in Lectures 6 and 8.

g- You are now considering an alternative CPM loan for the same amount and with the same interest rate and additional charges discussed above at points -b and -d. However, you decide to cap payments at £1,900. How much do you need to reimburse to the bank if you hold the mortgage to maturity? And what is the effective cost of borrowing of this loan? Comment on your results (15 points)

This a simple calculation that requires setting the PMT to 1900, and solving for FV instead of solving for PMT. This gives a final outstanding balance of 389,217.62 – and as clear from the figure the loan has a negative amortising feature. This is because 1900 is not enough to cover the cost of principal (10%/12 *250000 per month = 2083.33. So the missed interest payments and the non-amortization are compounding. The IRR of the loan can be found with 247500 (the loan with discount points), 239 PMT of 1900 and one final PMT of 1900 and 389217.64 = 10.10%. This has a slightly lower cost than the CPM loan with discount points above as more of the expensive charges are pushed in the future, and payments earlier on (that matters more in present discounted value) are smaller.

h- What is the effective cost of borrowing if instead you reimburse the loan after 10 years? And after 5 years? (10 points)

The students have to work out FV after 5 or 10 years, setting PMT to 1900 and the other mortgage quantities as instructed. This gives an outstanding balance after 5 years of 264,197 and after 10 years of 287,555. Notice that these are smaller than if one holds to maturity because the missed amortization and interest charges are not compounding. The IRR at 5 years and 10 years need to consider that the amount borrowed is 247500 and that the outstanding balances have a 2% penalty making them 269481 and 293306.

For 10 years, this gives an IRR of 10.279% (this is 119 periods of 1900 and 1 period of 1900 + 293306 vis a vis a loan of 247,500). For 5 years, this gives an IRR of 10.568% (this is 59 periods of 1900 and 1 period of 1900 + 269481 vis a vis a loan of 247,500). The results are expected – the sooner you repay, the higher the cost of borrowing with discount points and fees.

i- Consider now an alternative adjustable-rate mortgage (ARM) with monthly payments and compounding, and initially structured over 20 years. The mortgage charges an 8% annual interest rate for the first two years – before resetting at the beginning of Year 3 and again at the beginning of Year 5. You expect the annual interest rate you will be charged at the beginning of Year 3 to be 10%, before a further climb in Year 5 to 12%. What are the payments for this loan in the three periods when different rates prevail? What is the effective cost of borrowing if you hold this loan to maturity? (15 points)

ARM loans can be considered as sequences of CPM loans. The first loan would be for the full 20 years, charging 8% on the full loan of 250,000 with monthly compounding. This would give a payment of 2091. After two years (24 periods), using the AMORT worksheet students can find the outstanding loan is 238,993. This is the new principal for the loan charging 10% from the beginning of year 3 for the remaining 18 years or 216 periods. The new CPM at 10% for 18 years on 238993 gives a PMT of 2389.56. After two years (24 periods), the outstanding balance (with the AMORT function) of this loan would be 228468.5. This is now the new principal for the loan charging 12% for 16 years. The payment is 2681.59. The IRR of this loan can only be found with Cash Flow worksheet. This requires a loan of 247,500, then 24 payments of 2091. Then 24 payments of 2389.56 and finally 192 payments of 2681.59. This gives an IRR of 10.5679%. Expectedly, this is close to 10% - the interest charge of the central part of the loan – as future payments (at 12%) are less important in present terms.

j- You are now considering the ARM at point i- but would like to cap the payments for the first two years to £1,900. What would be the new payments after the interest rate resets in Year 3 and then again in Year 5? What is the effective cost of borrowing of this loan if you hold it to maturity? How does it compare to the one at point i-? (15 points)

The first thing students need to do is to work out the FV – i.e. the outstanding loan balance – after two years of paying 1900. This cannot be found with the AMORT worksheet. This is 243,948.9. Using this, students can calculate the new loan for 18 years charging 10% on this amount. That gives a payment of 2439.11 and an outstanding balance after two years of 233,206.16. Finally, the latter can be used as the new value of the last part of the loan charging 12% over 16 years. This gives a payment of 2737.199. The effective cost of borrowing then becomes 10.689%. This is slightly larger than before because the principal on which the 12% interest charges are applied is now larger because of the partial amortization feature of the initial part of the loan.